Linear Functions & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line $6x + y = 1$?

Answer: -6. Solve for y: $y = -\frac{6}{1}x + \frac{1}{1}$, so the slope is -6.

2 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = 7x - 11$, what is $f(-9)$?

Answer: -74. $f(-9) = 7x - 11|_{x=-9} = 7(-9) - 11 = -74$

3 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line has slope -8 and passes through the point $(3, -9)$. What is the y-intercept of the line?

Answer: 15. $y = mx + b \Rightarrow -9 = -8(3) + b \Rightarrow b = 15$

4 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line has slope -1 and passes through the point $(1, -3)$. What is the y-intercept of the line?

Answer: -2. $y = mx + b \Rightarrow -3 = -1(1) + b \Rightarrow b = -2$

5 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The table shows that $f(-1) = -8$ and $f(1) = 2$. Assuming f is linear, what is $f(3)$?

Answer: 12. Slope $= \frac{2+8}{1+1} = 5$. $f(x) = 5x - 3$. $f(3) = 12$.

6 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = -7x - 4$, what is $f(6)$?

Answer: -46. $f(6) = -7x - 4|_{x=6} = -7(6) - 4 = -46$

7 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line is parallel to the line $y = -4x + 3$ and passes through the point $(-7, -5)$. What is its y-intercept?

Answer: -33. Parallel to lines have slope -4. $-5 = -4(-7) + b \Rightarrow b = -33$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

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8 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The table shows that $f(1) = 10$ and $f(3) = 20$. Assuming f is linear, what is $f(5)$?

Answer: 30. Slope $= \frac{20-10}{3-1} = 5$. $f(x) = 5x + 5$. $f(5) = 30$.

9 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line $3x + y = 17$?

Answer: -3. Solve for y: $y = -\frac{3}{1}x + \frac{17}{1}$, so the slope is -3.

10 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The point $(4, 14)$ lies on the line $y = kx - 2$ in the xy-plane, where k is a constant. What is the value of k?

Answer: 4. Substitute $x = 4$ and $y = 14$: $14 = 4k - 2 \Rightarrow 4k = 16 \Rightarrow k = 4$.

11 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = 7x + 6$, what is $f(3)$?

Answer: 27. $f(3) = 7x + 6|_{x=3} = 7(3) + 6 = 27$

12 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line has slope -7 and passes through the point $(-2, 12)$. What is the y-intercept of the line?

Answer: -2. $y = mx + b \Rightarrow 12 = -7(-2) + b \Rightarrow b = -2$

13 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = x + 12$, what is $f(-8)$?

Answer: 4. $f(-8) = x + 12|_{x=-8} = 1(-8) + 12 = 4$

14 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = -7x + 4$, what is $f(2)$?

Answer: -10. $f(2) = -7x + 4|_{x=2} = -7(2) + 4 = -10$

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line in the xy-plane that passes through the points $(-3, 5)$ and $(1, -3)$?

Answer: -2. Slope $m = \frac{-3-5}{1-(-3)} = \frac{-8}{4} = -2$.

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16 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the horizontal line defined by $y = 7$ in the xy-plane?

Answer: 0. A horizontal line has zero vertical change ($\Delta y = 0$), so its slope is 0.

17 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The table shows that $f(-4) = 2$ and $f(-2) = -4$. Assuming f is linear, what is $f(0)$?

Answer: -10. Slope $= \frac{-4-2}{-2+4} = -3$. $f(x) = -3x - 10$. $f(0) = -10$.

18 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = -7x - 11$, what is $f(-3)$?

Answer: 10. $f(-3) = -7x - 11|_{x=-3} = -7(-3) - 11 = 10$

19 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line passes through the points $(-1, 15)$ and $(3, 7)$. If $f(x)$ represents this line, what is $f(-6)$?

Answer: 25. Slope $= \frac{7-15}{3+1} = -2$. $f(x) = -2x + 13$. $f(-6) = 25$.

20 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line passes through the points $(2, 4)$ and $(-4, 10)$. If $f(x)$ represents this line, what is $f(-5)$?

Answer: 11. Slope $= \frac{10-4}{-4-2} = -1$. $f(x) = -x + 6$. $f(-5) = 11$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which values of x satisfy $-2x + 3 > 7$?

a. $x > -2$

b. $x < -2$ ✓

c. $x > 2$

d. $x < 2$

Applying Linear Functions & Linear Inequalities principles, $-2x > 4 \Rightarrow x < -2$ Therefore, the verified correct option is $x < -2$.

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22 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of a function is shown. At which x-value does the function equal zero?

a. $x = -2$

b. $x = 0$

c. $x = 1$

d. $x = 2$ ✓

The zero (x-intercept) is where the graph crosses the x-axis: $x = 2$.

23 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line in the xy-plane has a slope of 3 and passes through the point $(2, 11)$. Which equation represents this line?

a. $y = 3x + 5$ ✓

b. $y = 3x + 11$

c. $y = 3x - 5$

d. $y = 2x + 3$

Using point-slope form: $y - 11 = 3(x - 2) \Rightarrow y = 3x - 6 + 11 = 3x + 5$.

24 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which of the following equations represents a horizontal line passing through the point $(3, -5)$?

a. $y = -5$ ✓

b. $x = 3$

c. $y = 3x - 5$

d. $y = -5x + 3$

A horizontal line has slope 0 and an equation of the form $y = c$. Since it passes through $(3, -5)$, the equation is $y = -5$.

25 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A gym charges a monthly membership fee of USD 30 plus USD 2 for each fitness class attended. If a member attends c fitness classes in a month, which expression represents the member's total cost for that month?

a. $30 + 2c$ ✓

b. $30c + 2$

c. $2(30 + c)$

d. $\frac{30 + c}{2}$

The base monthly fee is 30 and the variable fee is $2c$, giving $30 + 2c$.

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26 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line has equation $y = -4x + 9$. What is the y-intercept of the line?

a. $(0, -4)$

b. $(0, 9)$ ✓

c. $(9, 0)$

d. $(-4, 0)$

In slope-intercept form $y = mx + b$, the y-intercept is $(0, b) = (0, 9)$.

27 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which inequality represents all values of x such that $4x - 5 \leq 19$?

a. $x \leq 4$

b. $x \leq 5$

c. $x \leq 6$ ✓

d. $x \leq 7$

Add 5: $4x \leq 24$. Divide by 4: $x \leq 6$.

28 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line $2x + 5y = 20$?

a. $-\frac{2}{5}$ ✓

b. $\frac{2}{5}$

c. $-\frac{5}{2}$

d. 4

Solve for y: $5y = -2x + 20 \Rightarrow y = -\frac{2}{5}x + 4$. The slope is $-\frac{2}{5}$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Sarah earns USD 15/hour working h hours plus a USD 50 weekly bonus. Which expression is her weekly earnings?

a. $15 + 50h$

b. $15h + 50$ ✓

c. $50h + 15$

d. $65h$

The base earnings for h hours at USD 15 per hour is $15h$. Adding the fixed bonus of USD 50 yields total weekly earnings of $15h + 50$.

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30 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A taxi company charges an initial booking fee plus a fixed rate per mile driven. A trip of 4 miles costs USD 14, and a trip of 8 miles costs USD 22. What is the initial booking fee?

a. USD 4

b. **USD 6** ✓

c. USD 8

d. USD 2

Rate per mile: $m = \frac{22-14}{8-4} = \frac{8}{4} = 2$ dollars/mile. Initial fee: $14 - 2(4) = 14 - 8 = 6$ dollars.

31 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which of the following lines is parallel to the line $y = 4x - 9$ in the xy-plane?

a. $y = 4x + 12$ ✓

b. $y = -4x - 9$

c. $y = -\frac{1}{4}x + 3$

d. $y = \frac{1}{4}x - 9$

Parallel lines have identical slopes. The line $y = 4x - 9$ has slope 4, so $y = 4x + 12$ is parallel to it.

32 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which value of x satisfies the inequality $-3x + 8 < -7$?

a. 3

b. 4

c. 5

d. 6 ✓

Subtract 8: $-3x < -15$. Divide by -3 (reverse the inequality): $x > 5$. Among the choices, only 6 is strictly greater than 5.

33 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A plumber's total charge in dollars for a job lasting h hours is given by $C(h) = 31h + 25$. What does the number 25 represent in this context?

a. **the flat fee charged even if no labor hours are worked** ✓

b. the number of hours the plumber worked

c. the hourly rate the plumber charges

d. the total charge for a 1-hour job

In $C(h) = mh + b$, the constant term $b = 25$ is the value of C when $h = 0$: the flat fee.

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34 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A baker can produce at most 60 pastries each morning. If c represents the number of croissants and m represents the number of muffins, which inequality represents this restriction?

a. $c + m \leq 60$ ✓

b. $c + m \geq 60$

c. $cm \leq 60$

d. $c \leq 60m$

'At most 60' means the sum of croissants and muffins is less than or equal to 60: $c + m \leq 60$.

35 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which of the following points lies in the region defined by $y \geq x + 5$ in the xy-plane?

a. $(1, 8)$ ✓

b. $(2, 5)$

c. $(4, 7)$

d. $(3, 0)$

Test $(1, 8)$: $8 \geq 1 + 5 = 6$ (True). All other options fail: $5 < 7$, $7 < 9$, $0 < 8$.

36 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which of the following inequalities is equivalent to $4x - 6y > 18$?

a. $y < \frac{2}{3}x - 3$ ✓

b. $y > \frac{2}{3}x - 3$

c. $y < -\frac{2}{3}x + 3$

d. $y > -\frac{2}{3}x - 3$

$-6y > -4x + 18$. Dividing both sides by -6 reverses the inequality sign: $y < \frac{-4}{-6}x + \frac{18}{-6} \Rightarrow y < \frac{2}{3}x - 3$.

37 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which point satisfies the inequality $y < x + 5$?

a. (-1, 4)

b. (-5, 5)

c. (5, 14)

d. **(-5, -3)** ✓

Substitute each point; only $(-5, -3)$ makes $y < 1x + 5$ true.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the y-intercept of the line $5x - 2y = 20$ in the xy-plane?

a. $(0, -10)$ ✓

b. $(0, 10)$

c. $(4, 0)$

d. $(-4, 0)$

To find the y-intercept, set $x = 0$: $5(0) - 2y = 20 \Rightarrow -2y = 20 \Rightarrow y = -10$. Thus, the y-intercept is $(0, -10)$.

39 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

At what point does the line $y = -\frac{1}{2}x + 4$ intersect the x-axis in the xy-plane?

a. $(8, 0)$ ✓

b. $(-8, 0)$

c. $(0, 4)$

d. $(4, 0)$

To find the x-intercept, set $y = 0$: $0 = -\frac{1}{2}x + 4 \Rightarrow \frac{1}{2}x = 4 \Rightarrow x = 8$. The point is $(8, 0)$.

40 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of any line in the xy-plane that is perpendicular to the line $y = -\frac{2}{5}x + 3$?

a. $\frac{5}{2}$ ✓

b. $-\frac{5}{2}$

c. $\frac{2}{5}$

d. $-\frac{2}{5}$

The slope of the given line is $-\frac{2}{5}$. Perpendicular lines have negative reciprocal slopes: $-\frac{1}{-2/5} = \frac{5}{2}$.

41 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line passing through $(2, 5)$ and $(6, 17)$?

a. 2

b. 3 ✓

c. 4

d. 5

$m = \frac{17-5}{6-2} = \frac{12}{4} = 3$.

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42 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The function $f(x) = 15x + 40$ models the total charge in dollars for x hours of equipment rental. What does the value 40 represent?

a. The hourly rate for the rental

b. **The flat initial fee charged regardless of hours ✓**

c. The maximum number of rental hours

d. The total charge for 15 hours

When $x = 0$, $f(0) = 40$. Thus, 40 represents the fixed initial fee.

43 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A plumber's total charge in dollars for a job lasting h hours is given by $C(h) = 51h + 41$. What does the number 41 represent in this context?

a. the total charge for a 1-hour job

b. **the flat fee charged even if no labor hours are worked ✓**

c. the number of hours the plumber worked

d. the hourly rate the plumber charges

In $C(h) = mh + b$, the constant term $b = 41$ is the value of C when $h = 0$: the flat fee.

44 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which point satisfies the inequality $y < 2x - 2$?

a. (-2, 0)

b. (-5, -7)

c. (-4, -10)

d. **(-2, -9) ✓**

Substitute each point; only $(-2, -9)$ makes $y < 2x - 2$ true.

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45 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A region in the xy-plane is bounded by a dashed line with y-intercept $(0, 3)$ and slope -2, and the region below this line is shaded. Which inequality represents this region?

a. $y < -2x + 3$ ✓

b. $y \leq -2x + 3$

c. $y > -2x + 3$

d. $y \geq -2x + 3$

A dashed boundary indicates a strict inequality ($<$ or $>$), and shading below corresponds to $<$. Thus, $y < -2x + 3$.

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